Data Wrangling (Data Preprocessing)

Code ▾

Practical assessment 1

Prathibha M

19.08.2024

Setup

Hide

```
# **Packages**

library(kableExtra)
library(magrittr)
```

STEP 1: Student Detail

Group information

Student Name	Student Number	Contribution percentage
Prathibha M	s3859590	100%

STEP 2: Description of Dataset

- Dataset: “Students Performance in Exams” from Kaggle
- Source: Kaggle Dataset
- URL: https://www.kaggle.com/datasets/spscientist/students-performance-in-exams/data?select=StudentsPerformance.csv (https://www.kaggle.com/datasets/spscientist/students-performance-in-exams/data?select=StudentsPerformance.csv)
- Features : contains 1,000 rows (entries) and 8 columns (features)

- 8 Columns are:
- 1.gender: The gender of the student (male/female).
- 2.race/ethnicity: The student’s race or ethnicity, categorized into five groups (Group A, B, C, D, E).
- 3.parental level of education: The highest level of education attained by the student’s parents, ranging from high school to master’s degree.
- 4.lunch: The type of lunch the student receives (standard or free/reduced).
- 5.test preparation course: Whether the student completed a test preparation course (none/completed).
- 6.math score: The student’s score in mathematics (0-100).
- 7.reading score: The student’s score in reading (0-100).
- 8.writing score: The student’s score in writing (0-100).
- Numeric Variables: math score, reading score, writing score
- Categorical Variables: gender, race/ethnicity, parental level of education, lunch, test preparation course
- The dataset provides both numeric and categorical data about student performance and background, making it suitable for analysis.

STEP 3: Importing Data

Hide

```
# Read the data
dt <- read.csv("StudentsPerformance.csv", header = TRUE)

#structure of the data
str(dt)
```

```
'data.frame':	1000 obs. of  8 variables:
 $ gender                     : chr  "female" "female" "female" "male" ...
 $ race.ethnicity             : chr  "group B" "group C" "group B" "group A" ...
 $ parental.level.of.education: chr  "bachelor's degree" "some college" "master's degree" "associate's degree" ...
 $ lunch                      : chr  "standard" "standard" "standard" "free/reduced" ...
 $ test.preparation.course    : chr  "none" "completed" "none" "none" ...
 $ math.score                 : int  72 69 90 47 76 71 88 40 64 38 ...
 $ reading.score              : int  72 90 95 57 78 83 95 43 64 60 ...
 $ writing.score              : int  74 88 93 44 75 78 92 39 67 50 ...
```

Hide

```
# Display the first few rows of the dataset
head(dt)
```

	gender <chr>	race.ethnicity <chr>	parental.level.of.education <chr>	lunch <chr>	test.preparation.course <chr>
1	female	group B	bachelor's degree	standard	none
2	female	group C	some college	standard	completed
3	female	group B	master's degree	standard	none
4	male	group A	associate's degree	free/reduced	none
5	male	group C	some college	standard	none
6	female	group B	associate's degree	standard	none

6 rows | 1-6 of 8 columns

Hide

```
# Check for missing values
summary(dt)
```

```
   gender          race.ethnicity     parental.level.of.education    lunch            test.preparation.course   math.score
 Length:1000        Length:1000        Length:1000                 Length:1000        Length:1000              Min.   :  0.00
 Class :character   Class :character   Class :character            Class :character   Class :character         1st Qu.: 57.00
 Mode  :character   Mode  :character   Mode  :character            Mode  :character   Mode  :character         Median : 66.00
                                                                                                               Mean   : 66.09
                                                                                                               3rd Qu.: 77.00
                                                                                                               Max.   :100.00
 reading.score    writing.score
 Min.   : 17.00   Min.   : 10.00
 1st Qu.: 59.00   1st Qu.: 57.75
 Median : 70.00   Median : 69.00
 Mean   : 69.17   Mean   : 68.05
 3rd Qu.: 79.00   3rd Qu.: 79.00
 Max.   :100.00   Max.   :100.00
```

- The dataset is imported using read.csv() function,loads Csv file into R dataframe.
- The dataset structure is viewed with str()
- The head(dt) function is used to view the first few rows of the dataset
- A statistical summary of each column in the dataframe is shown using summary()

STEP 4: Inspect and Understand

Hide

```
# Check dimensions
dim(dt)
```

```
[1] 1000    8
```

Hide

```
# Check the column names
colnames(dt)
```

```
[1] "gender"                      "race.ethnicity"              "parental.level.of.education" "lunch"
[5] "test.preparation.course"     "math.score"                  "reading.score"               "writing.score"
```

Hide

```
# Rename columns and check column names
colnames(dt) <- c("gen", "r_eth", "pa_ed", "lun", "t_prep", "m_scr", "r_scr", "w_scr")
colnames(dt)
```

```
[1] "gen"    "r_eth"  "pa_ed"  "lun"    "t_prep" "m_scr"  "r_scr"  "w_scr"
```

Hide

```
# structure of the dataset
str(dt)
```

```
'data.frame':   1000 obs. of  8 variables:
 $ gen   : chr  "female" "female" "female" "male" ...
 $ r_eth : chr  "group B" "group C" "group B" "group A" ...
 $ pa_ed : chr  "bachelor's degree" "some college" "master's degree" "associate's degree" ...
 $ lun   : chr  "standard" "standard" "standard" "free/reduced" ...
 $ t_prep: chr  "none" "completed" "none" "none" ...
 $ m_scr : int  72 69 90 47 76 71 88 40 64 38 ...
 $ r_scr : int  72 90 95 57 78 83 95 43 64 60 ...
 $ w_scr : int  74 88 93 44 75 78 92 39 67 50 ...
```

Hide

```
# Convert appropriate columns to factor
dt$gen <- as.factor(dt$gen)
dt$r_eth <- as.factor(dt$r_eth)
dt$pa_ed <- as.factor(dt$pa_ed)
dt$lun <- as.factor(dt$lun)
dt$t_prep <- as.factor(dt$t_prep)

# Check levels of factor
levels(dt$gen)
```

```
[1] "female" "male"
```

Hide

```
levels(dt$r_eth)
```

```
[1] "group A" "group B" "group C" "group D" "group E"
```

Hide

```
levels(dt$pa_ed)
```

```
[1] "associate's degree" "bachelor's degree"  "high school"        "master's degree"    "some college"       "some high scho
ol"
```

Hide

```
levels(dt$lun)
```

```
[1] "free/reduced" "standard"
```

Hide

```
levels(dt$t_prep)
```

```
[1] "completed" "none"
```

Hide

```
#renaming levels
dt$pa_ed <- factor(dt$pa_ed, levels = c("some high schl", "high schl", "asso degree", "bch degree", "master degree"))
levels(dt$pa_ed)
```

```
[1] "some high schl" "high schl"      "asso degree"    "bch degree"     "master degree"
```

Hide

```
# Summary
summary(dt)
```

```
     gen           r_eth                pa_ed                lun             t_prep          m_scr            r_scr       
 female:518   group A: 89   some high schl:   0   free/reduced:355   completed:358   Min.   :  0.00   Min.   : 17.00  
 male  :482   group B:190   high schl     :   0   standard    :645   none     :642   1st Qu.: 57.00   1st Qu.: 59.00  
              group C:319   asso degree   :   0                                      Median : 66.00   Median : 70.00  
              group D:262   bch degree    :   0                                      Mean   : 66.09   Mean   : 69.17  
              group E:140   master degree :   0                                      3rd Qu.: 77.00   3rd Qu.: 79.00  
                            NA's          :1000                                      Max.   :100.00   Max.   :100.00  
     w_scr       
 Min.   : 10.00  
 1st Qu.: 57.75  
 Median : 69.00  
 Mean   : 68.05  
 3rd Qu.: 79.00  
 Max.   :100.00  
```

- I used dim() to see the size/dimensions of dataset.
- I used colnames() shows names of the columns. Then I updated the column names using a vector of new names and check the changes.
- The dataset structure is viewed with str(),ensuring the data types were correct.
- Categorical variables were converted using as.factor(), and factor levels were check levels() and adjusted as needed.
- To redefine and order the factor level factor() is used.
- A statistical summary of each column in the dataframe is shown using summary()

STEP 5: Subsetting

Hide

```
# Subset the data to include first 10 obs
ss_dt <- dt[1:10, ]

# Convert subset to a matrix
ss_matrix <- as.matrix(ss_dt)

# Check structure of matrix
str(ss_matrix)
```

```
chr [1:10, 1:8] "female" "female" "female" "male" "male" "female" "female" "male" "male" "female" "group B" "group C" ...
- attr(*, "dimnames")=List of 2
 ..$ : chr [1:10] "1" "2" "3" "4" ...
 ..$ : chr [1:8] "gen" "r_eth" "pa_ed" "lun" ...
```

- I subsetted the first 10 rows of the dataset and converted them to a matrix and checking the structure with str().
- The matrix contains only one data type (character) because matrices in R can only store a single type of data, so mixed types (numeric, factor) in the original data frame are converted to character.

STEP 6: Create a new Data Frame

Hide

```r
##Create Integer and Ordinal Variables
inte_v <- 1:10
ordnl_v <- factor(c("Low", "Medium", "High", "Medium", "Low", "High", "Medium", "Low", "High", "Medium"),
                       levels = c("Low", "Medium", "High"), ordered = TRUE)

## Combine Variables into Data Frame
df <- data.frame(Intev = inte_v, Ordnlv = ordnl_v)

# Display the structure of new data frame
str(df)
```

```
'data.frame':	10 obs. of  2 variables:
 $ Intev : int  1 2 3 4 5 6 7 8 9 10
 $ Ordnlv: Ord.factor w/ 3 levels "Low"<"Medium"<..: 1 2 3 2 1 3 2 1 3 2
```

Hide

```r
# Create numeric vector
adtnl_v <- c(10, 20, 30, 40, 50, 60, 70, 80, 90, 100)

# Add numeric vector as new column to existing data frame
df <- cbind(df, Adtnlv = adtnl_v)

##Check Updated Data Frame Structure
str(df)
```

```
'data.frame':	10 obs. of  3 variables:
 $ Intev : int  1 2 3 4 5 6 7 8 9 10
 $ Ordnlv: Ord.factor w/ 3 levels "Low"<"Medium"<..: 1 2 3 2 1 3 2 1 3 2
 $ Adtnlv: num  10 20 30 40 50 60 70 80 90 100
```

- I have created two variables: an integer vector from 1 to 10 and an ordinal factor with levels “Low”, “Medium”, and “High” to ensure correct ordering.
- I combined the integer and ordinal variables into a new data frame for structured analysis.

- Then a numeric vector is created for additional data and added this vector as a new column to the data frame.
- And to confirm the new column has been added, the updated structure of the data frame is displayed.